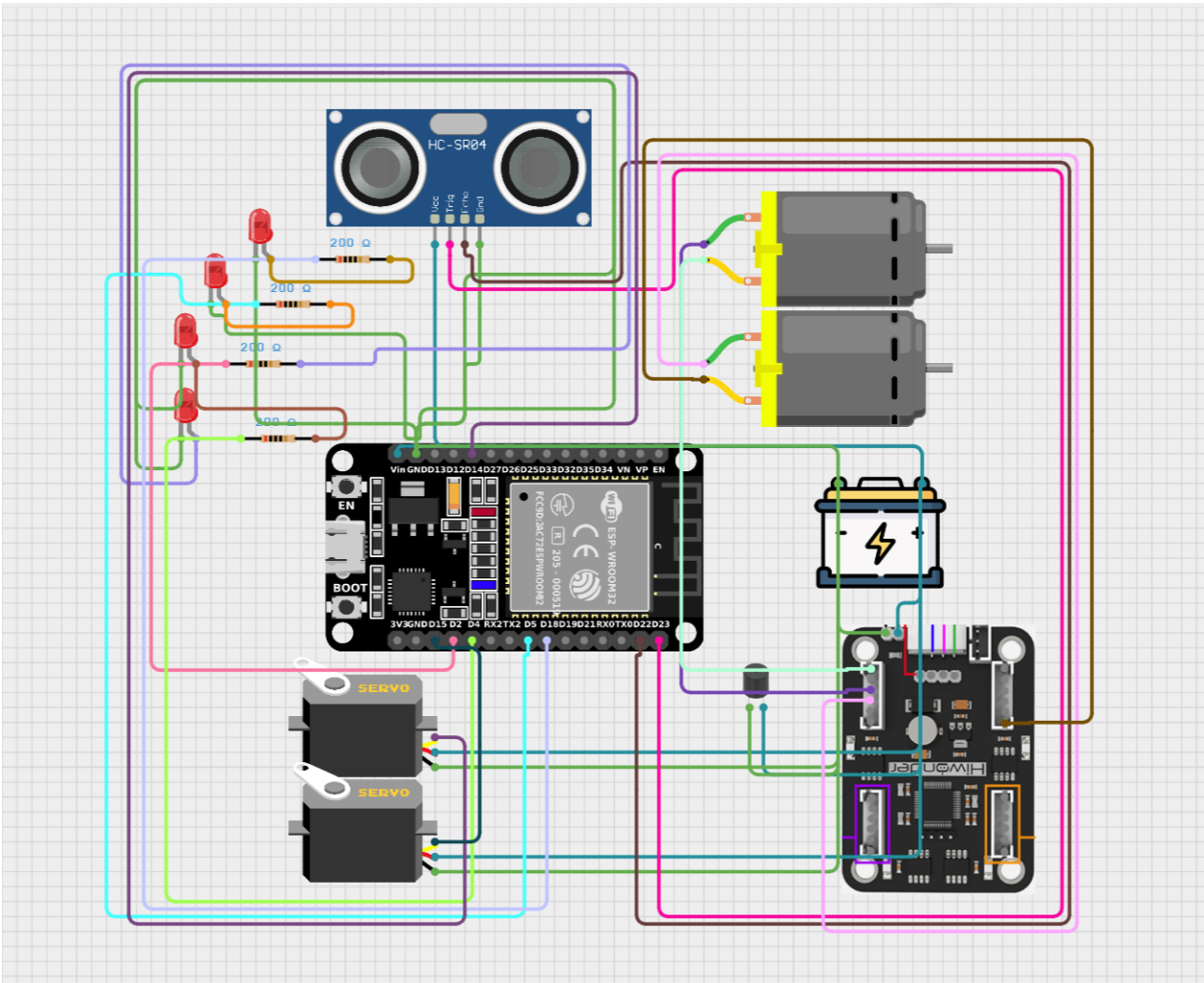
Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
1. LEDs	*4*	*Acts as the indicator for the car*	*Easy to code and create patterns to signal left and right*

2. Ultrasound sensor	1	To detect the presence of an obstacle in front	Gives more accurate result than capacitive touch and can be used to calculate stopping distance.
3. Buzzer	1	To indicate presence of an obstacle	Acts as an alarm to prevent car crash.
4. DC geared motors	2	To move the wheels forward, back, turn left and right	Smoother movements and precise speed control.
5. Servo Motors	2	To give the car an animatronic feel.	Compact in size and precise control.
6. ESP32	1	Wifi Control	Microprocessor to bring all the components together
7. Resistors	4	To complete the circuiting for LEDs	Controls the voltage and current input for LEDs
8. Battery (LiPo)	1	Power supply	Compact and produces enough voltage to run the car
9. Motor driver module	1	Direction control	Needs more power than just ESP32 and helps control motion
10. Breadboard	1	Connections	Convenient to keep all the components connected
11. Copper wires	As required	Connections	Convenient to keep all the components connected
12. MIT app inventor	1	Remote control	Easily portable and compatible remote control system